Red Teaming the Mind of the Machine: A Systematic Evaluation of Prompt Injection and Jailbreak Vulnerabilities in LLMs

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arXiv:2505.04806v2 [cs.CR] 13 May 2025

***Abstract*—Large Language Models (LLMs) are increasingly integrated into consumer and enterprise applications. Despite their capabilities, they remain susceptible to adversarial attacks such as prompt injection and jailbreaks that override alignment safeguards. This paper provides a systematic investigation of jailbreak strategies against various state-of-the-art LLMs. We categorize over 1,400 adversarial prompts, analyze their success against GPT-4, Claude 2, Mistral 7B, and Vicuna, and examine their generalizability and construction logic. We further propose layered mitigation strategies and recommend a hybrid red-teaming and sandboxing approach for robust LLM security.**

***Index Terms*—Large Language Models, Prompt Injection, Jailbreak, Adversarial Prompts, AI Security, Red Teaming, LLM Safety**

I. Introduction

The field of artificial intelligence has experienced a paradigm shift with the emergence of large language models (LLMs). These systems have transitioned from research prototypes to core components of production-grade systems, shaping industries from finance and law to healthcare and entertainment. LLMs are praised for their fluency, contextual reasoning, and ability to generate human-like responses. However, these capabilities also expose them to a new class of security threats. As LLMs are increasingly used in decision-making systems, chatbots, content moderation tools, and virtual agents, the potential for abuse through adversarial inputs grows exponentially.

Large Language Models (LLMs) have fundamentally transformed the landscape of natural language processing, enabling applications in content generation, customer service, coding assistance, legal analysis, and more. With models like OpenAI's GPT-4 [17], Anthropic's Claude 2 [18], Meta's LLaMA [50], and open-source offerings such as Vicuna [51] and Mistral 7B [52], LLMs now influence millions of users globally. However, this ubiquity introduces significant security concerns, particularly surrounding adversarial prompt engineering techniques that manipulate model behavior. These techniques, often referred to as prompt injection or jailbreaks, are capable of bypassing built-in safety filters and elicit outputs that violate platform policies, such as generating hate speech, misinformation, or malicious code [1] [2] [4].

Prompt injection represents a new class of vulnerabilities unique to LLMs. Unlike traditional software vulnerabilities rooted in memory safety or access control flaws, prompt injection leverages the interpretive nature of natural language inputs [5] [6]. This paper explores the mechanisms and success of prompt injection across a range of LLMs, documenting the systemic weaknesses that attackers exploit [2] [7] [8].
Contributions of this work include:

- A comprehensive taxonomy of jailbreak prompts categorized by attack vector [3] [6]
- Empirical evaluation of prompt effectiveness across closed and open-source LLMs [4] [10] [11]
- Scenario-specific attack success analysis in domains such as law, politics, and security [1] [9] [13]
- Discussion of community-based jailbreak dissemination and its parallels to exploit markets [14] [15] [27]
- Detailed recommendations for mitigating prompt injection vulnerabilities [7] [12] [16]

II. Background

A. Overview of Large Language Models

Large Language Models operate using billions of parameters and are trained on diverse datasets encompassing text from books, articles, websites, and code. Notable models such as GPT-4, Claude 2, and Mistral 7B build on earlier architectures but have significantly improved reasoning, factual recall, and stylistic flexibility. The ability of these models to learn from few examples, a phenomenon called in-context learning, contributes to their versatility but also to their vulnerability. When exposed to crafted prompts, these models can be misled into misaligned behavior.

LLMs are built on the transformer architecture introduced by Vaswani et al. in 2017 [47]. Recent advancements include autoregressive pretraining on massive corpora followed by supervised finetuning and alignment through Reinforcement Learning from Human Feedback (RLHF) [34].

B. Alignment and Safety Mechanisms

To prevent harmful output, LLMs rely on several safety mechanisms, including instruction tuning [28], reinforcement from rejection sampling (RLAIF), pre- and post-output moderation filters [16], and system prompts embedding safety guidelines [12] [33].

攻击机器思维：对 LLMs 中提示注入和越狱漏洞的系统评估

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arXiv:2505.04806v2 [cs.CR] 13 May 2025

摘要—大型语言模型（LLMs）正越来越多地集成到消费者和企业应用中。尽管它们功能强大，但仍容易受到提示注入和越狱等对抗性攻击的威胁，这些攻击会绕过对齐安全措施。本文对针对各种最先进 LLMs 的越狱策略进行了系统研究。我们对 1400 多个对抗性提示进行了分类，分析了它们对 GPT-4、Claude 2、Mistral 7B 和 Vicuna 的成功率，并考察了它们的泛化能力和构建逻辑。我们进一步提出了分层缓解策略，并推荐了一种混合红队测试和沙箱方法的混合方法，以实现强大的 LLM 安全。

索引词—大型语言模型，提示注入，越狱，对抗性提示，人工智能安全，红队演练，LLM 安全

I. I

人工智能领域随着大型语言模型（LLMs）的出现发生了范式转变。这些系统已从研究原型转变为生产级系统的核心组件，塑造了从金融和法律到医疗保健和娱乐的行业。LLMs 因其流畅性、情境推理能力以及生成类人响应的能力而受到赞誉。然而，这些能力也使它们暴露于新的安全威胁类别中。随着 LLMs 越来越多地用于决策系统、聊天机器人、内容审核工具和虚拟代理，通过对抗性输入进行滥用的可能性呈指数级增长。

大型语言模型（LLMs）从根本上改变了自然语言处理领域，使内容生成、客户服务、编程辅助、法律分析等应用成为可能。以 OpenAI 的 GPT-4 [17]、Anthropic 的 Claude 2 [18]、Meta 的 LLaMA [50]以及 Vicuna [51]和 Mistral 7B [52]等开源模型为例，LLMs 现在影响着全球数百万用户。然而，这种普遍性带来了重大的安全问题，特别是围绕对抗性提示工程技术的问题，这些技术能够操纵模型行为。这些技术通常被称为提示注入或越狱，能够绕过内置的安全过滤器，并产生违反平台政策的结果，例如生成仇恨言论、虚假信息或恶意代码 [1] [2] [4]。

提示注入代表了一种仅适用于 LLMs 的新型漏洞。与传统软件漏洞不同基于内存安全或访问控制缺陷，提示注入利用了自然语言输入的解释性特点[5][6]。本文探讨了提示注入在多种 LLMs 中的机制和成功率，记录了攻击者利用的系统弱点[2][7][8]。

本工作的贡献包括：

- 按攻击向量分类的越狱提示的综合分类法[3][6]
- 在闭源和开源 LLMs 中对提示有效性的实证评估[4][10][11]
- 特定场景下的攻击成功分析，涉及法律、政治和安全领域[1][9][13]
- 关于社区驱动的越狱传播及其与利用市场的相似性[14][15][27]
- 缓解提示注入漏洞的详细建议[7][12][16]

II. B

A. 大型语言模型概述

大型语言模型使用数十亿个参数，并在包含书籍、文章、网站和代码的多样化数据集上进行训练。GPT-4、Claude 2 和 Mistral 7B 等知名模型基于早期架构进行改进，但在推理能力、事实回忆和风格灵活性方面有显著提升。这些模型能够通过少量示例进行学习，这种现象被称为情境学习，这既增强了它们的通用性，也增加了它们的脆弱性。当暴露于精心设计的提示时，这些模型可能会被误导而表现出不一致的行为。

LLMs 基于 Vaswani 等人在 2017 年提出的 Transformer 架构 [47]。最近的进展包括在大型语料库上进行自回归预训练，然后通过人类反馈强化学习（RLHF）进行监督微调和对齐 [34]。

B. 对齐和安全机制

为了防止有害输出，LLMs 依赖于多种安全机制，包括指令微调 [28]、拒绝采样的强化学习 (RLAIF)、输出前后的审核过滤器 [16]，以及嵌入安全指南的系统提示 [12] [33]。

C. Prompt Injection Explained

Prompt injection is the LLM analogue to command injection in traditional computing [1] [3] [6]. Common attack vectors include role-based conditioning, instruction hijacking, obfuscated encoding, and multi-turn manipulation [2] [5] [7]. Studies have shown that these attacks are reproducible, transferable, and can circumvent various filtering methods [3] [10] [14].

D. Related Work

Zhang et al. introduced a foundational taxonomy categorizing prompt injections [3]. Shen et al. aggregated 1,405 jailbreak prompts across 131 forums, revealing a 95% success rate in some cases [48]. Ding et al. developed ReNeLLM, which improved jailbreak performance by 40% [49]. Anthropic's many-shot prompt conditioning decreased attack success rates significantly [18]. OWASP's Top 10 identified prompt injection as the most critical vulnerability [16].
Other notable contributions include:

- Liu et al. on empirical jailbreak strategies [2]
- Yi et al. on indirect prompt injection detection [4]
- Suo et al. on defense techniques derived from attack insights [9]
- Chen et al. on preference-aligned defenses [11]
- William on bypass detection and Zhao et al. unified defenses [15] [13]
- Apurv et al. on threat modeling for red-teaming LLMs [53]

III. Methodology

This section outlines our approach to measuring LLM vulnerabilities with an emphasis on reproducibility and diversity. Our methodology integrates qualitative red teaming insights with quantitative metrics collected through structured prompt testing. All experiments are governed by ethical red-teaming principles. In addition to evaluating raw performance, we also tracked behavioral consistency and model self-awareness to adversarial stimuli. This dual-pronged framework allows us to detect subtle failure patterns beyond binary success metrics.

This section builds on emerging adversarial benchmarks such as JailbreakBench and RedBench, while drawing defense insights from frameworks like PromptShield [21] and Palisade [10]. We employed Sentence-BERT embeddings [46], GPT-based moderation strategies [17], and adversarial annotation heuristics from prior poisoning literature [22] to validate prompt behavior and misalignment tendencies.

A. Dataset Construction

We curated a dataset of 1,400+ adversarial prompts from:

- Public jailbreak repositories (e.g., GitHub, JailbreakChat)
- LLM exploit forums on Reddit and Discord
- Prior academic corpora, including JailbreakBench [3] and PromptBench [6]

Each prompt was manually validated, categorized into attack types (roleplay, logic traps, encoding, multi-turn), and annotated for content sensitivity (e.g., political, legal, explicit).

B. Target Models

We tested prompts on four models:

- GPT-4 (OpenAI, March 2024 snapshot)
- Claude 2 (Anthropic, July 2023 API version)
- Mistral 7B (open-weight model via Hugging Face Inference)
- Vicuna-13B (via local HF inference server)

Model versions were frozen to ensure reproducibility. All inference was performed using controlled prompts, with the system context initialized per the model's recommended safety guidelines.

C. Evaluation Metrics

We used the following primary metrics:

- Attack Success Rate (ASR): Whether the model produced an output violating its intended guardrails
- Prompt Generalizability: How often a prompt successful on one model succeeded on another
- Time-to-Bypass: Average minutes taken to successfully induce misaligned behavior
- Failure Mode Classification: Taxonomy of observed response behaviors (e.g., partial refusals, misleading responses)

D. Automation Pipeline

We developed a semi-automated red-teaming script using the LangChain framework. Prompts were injected via API calls (OpenAI, Claude) and local model inference (Mistral, Vicuna). Output was scored using a hybrid method:

- Keyword spotting (for trigger words)
- GPT-based meta-evaluation of harmfulness [17]
- Sentence-BERT semantic distance from refusal templates [46]

E. Defense Framework Evaluation

To simulate real-world defenses, we layered external filtering strategies on outputs:

- PromptShield ruleset [21]
- Palisade detection framework [10]
- Signed-Prompt verification logic [41]

We then retested a subset of successful jailbreaks against these defenses to estimate defense coverage and bypass rate.

IV. Results

The evaluation of prompt injection efficacy was conducted using a rigorous experimental design.

The figures, included in the appendix or digital supplement, visually illustrate comparative vulnerability trends. For instance, Figure 1 shows that GPT-4 exhibited the highest attack success rate. These results highlight not only model-specific weaknesses but also the effectiveness of specific prompt engineering tactics.

C. 提示注入解释

提示注入是传统计算中命令注入的 LLM 对应物[1][3][6]。常见的攻击向量包括基于角色的条件设置、指令劫持、混淆编码和多轮操纵[2][5][7]。研究表明，这些攻击是可重复的、可迁移的，并且可以绕过各种过滤方法[3][10][14]。

D. 相关工作

张等人引入了一个基础分类法来对提示注入进行分类[3]。沈等人汇总了 131 个论坛上的 1405 个越狱提示，揭示在某些情况下成功率高达 95%[48]。丁等人开发了 ReNeLLM，将越狱性能提高了 40%[49]。Anthropic 的许多样本提示条件显著降低了攻击成功率[18]。OWASP 的十大安全风险将提示注入识别为最关键的安全漏洞[16]。

其他值得注意的贡献包括：

- 刘等人关于实证越狱策略[2]
- 易等人关于间接提示注入检测[4]
- 苏等人关于基于攻击见解的防御技术[9]
- 陈等人关于偏好对齐的防御[11]
- 威廉关于绕过检测和赵等人统一防御[15][13]
- 阿普尔等人关于红队 LLM 威胁建模[53]

III. M 本节概述了我们测量 LLM 漏洞的方法，重点强调可重复性和多样性。我们的方法将定性的红队见解与通过结构化提示测试收集的定量指标相结合。所有实验均受道德红队原则的约束。除了评估原始性能外，我们还追踪了行为一致性和模型对对抗性刺激的自我意识。这种双管齐下的框架使我们能够检测到超越二元成功指标的微妙失败模式。

本节基于新兴的对抗基准，如 JailbreakBench 和 RedBench，同时从 PromptShield[21]和 Palisade[10]等框架中汲取防御见解。我们采用了 Sentence-BERT 嵌入[46]、基于 GPT 的审核策略[17]以及来自先前中毒文献的对抗性标注启发式方法[22]来验证提示行为和错位倾向。

A. 数据集构建

我们收集了一个包含 1,400 多条对抗性提示的数据集，来源包括：

- 公开的越狱仓库（例如，GitHub，JailbreakChat）
- Reddit 和 Discord 上的 LLM 漏洞利用论坛

先前的学术语料库，包括 JailbreakBench [3] 和 PromptBench [6]，每个提示都经过人工验证，按攻击类型（角色扮演、逻辑陷阱、编码、多轮）分类，并标注了内容敏感性（例如，政治、法律、露骨）。

B. 目标模型

我们在四个模型上测试了提示：

- GPT-4 (OpenAI, 2024 年 3 月快照)
- Claude 2 (Anthropic, 2023 年 7 月 API 版本)
- Mistral 7B (通过 Hugging Face 推理的开源模型)
- Vicuna-13B（通过本地 Hugging Face 推理服

模型版本已固定以确保可重复性。所有推理均使用受控提示进行，系统上下文根据模型推荐的安全指南初始化。

C. 评估指标

我们使用了以下主要指标：

- 攻击成功率（ASR）：模型是否生成了违反其预期约束条件的输出
- 提示泛化能力：一个提示在某个模型上成功后，在另一个模型上成功的频率
- 绕过时间：成功诱导行为偏差所需的平均分钟数
- 失败模式分类：观察到的响应行为分类（例如，部分拒绝、误导性响应）

D. 自动化流程

我们使用 LangChain 框架开发了一个半自动化的红队测试脚本。通过 API 调用（OpenAI、Claude）和本地模型推理（Mistral、Vicuna）注入提示。输出采用混合方法进行评分：

- 关键词识别（用于触发词）
- 基于 GPT 的有害性元评估[17]
- 与拒绝模板的 Sentence-BERT 语义距离[46]

E. 防御框架评估

为了模拟现实世界的防御措施，我们在输出上叠加了外部过滤策略：

- PromptShield 规则集 [21]
- Palisade 检测框架 [10]
- 签名提示验证逻辑 [41]

我们随后对成功逃逸的一小组案例针对这些防御措施进行了重新测试，以评估防御覆盖率和绕过率。

IV. R

提示注入的有效性评估是通过严格的实验设计进行的。

附录或数字补充中包含的图表直观地展示了比较脆弱性趋势。例如，图 1 显示 GPT-4 表现出最高的攻击成功率。这些结果不仅突出了模型特有的弱点，也显示了特定提示工程策略的有效性。

A. Model Susceptibility Analysis

Among the tested models, GPT-4 demonstrated the highest vulnerability with an ASR of 87.2%, confirming its powerful but permissive instruction-following nature. While Claude 2 performed slightly better in filtering, it still succumbed to 82.5% of attacks. Open models such as Mistral 7B (71.3%) and Vicuna (69.4%) revealed significant weaknesses, likely due to the absence of robust fine-tuned safety layers.

Interestingly, GPT-4 and Claude 2 shared structural similarities in moderation behavior-exhibiting soft refusals before ultimately yielding to adversarial logic, especially in legal, creative, or conditional prompts. These nuances suggest that model scale and alignment tuning complexity both contribute to attack surface depth.

In terms of generalizability, jailbreak prompts that succeeded on GPT-4 transferred effectively to Claude 2 and Vicuna in 64.1% and 59.7% of cases respectively.

Average time to generate a successful jailbreak was under 17 minutes for GPT-4, while Mistral required approximately 21.7 minutes on average. Our experiments evaluated over 1,400 adversarial prompts across four LLMs: GPT-4, Claude 2, Mistral 7B, and Vicuna. We analyze results along several dimensions, including model susceptibility, attack technique efficacy, prompt behavior patterns, and cross-model generalization.

Model	ASR (%)	Generalizability (%)	Time-to-Bypass (min)
GPT-4	87.2	64.1	16.2
Claude 2	82.5	59.7	17.4
Mistral 7B	71.3	52.4	21.7
Vicuna	69.4	50.6	20.9

TABLE I: Model-wise Evaluation Metrics

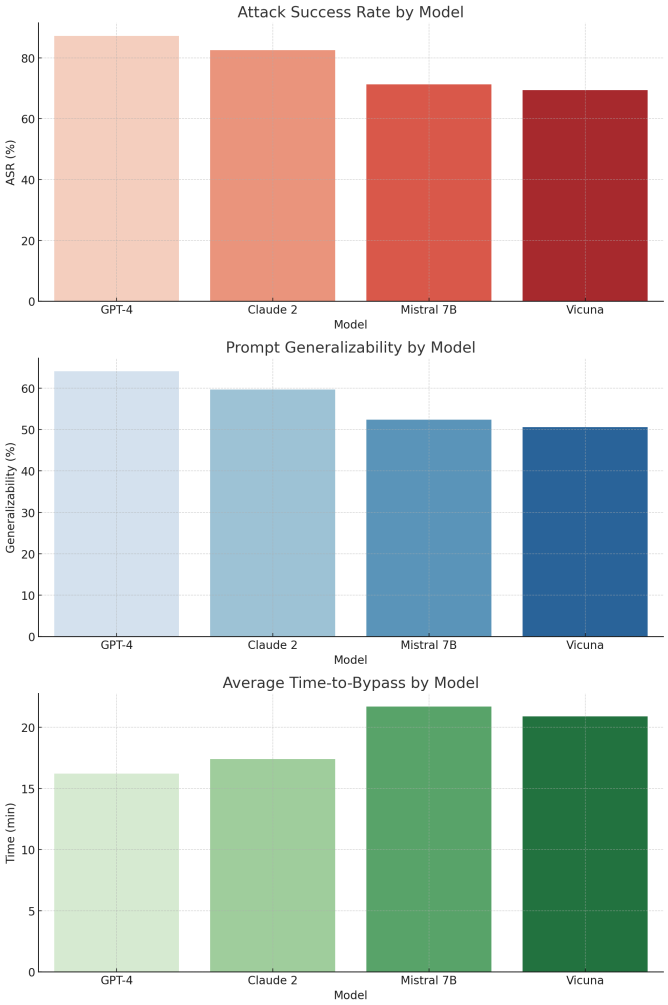


Fig. 1: Model-wise Evaluation Metrics

B. Attack Category Performance

Prompt injections exploiting roleplay dynamics (e.g., impersonation of fictional characters or hypothetical scenarios) achieved the highest ASR (89.6%). These prompts often bypass filters by deflecting responsibility away from the model (e.g., "as an AI in a movie script…").

Logic trap attacks (ASR: 81.4%) exploit conditional structures and moral dilemmas to elicit disallowed content. Encoding tricks (e.g., base64 or zero-width characters) achieved 76.2% ASR by evading keyword-based filtering mechanisms. While multi-turn dialogues yielded slightly lower effectiveness (68.7%), they often succeeded in long-form tasks where context buildup gradually weakened safety enforcement.

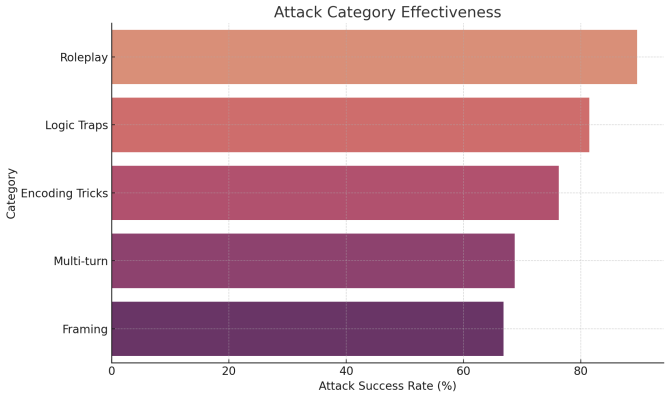


Fig. 2: Attack Category Effectiveness

C. Scenario-Specific Vulnerabilities

Targeted domains revealed non-uniform vulnerabilities:

- Political content: Prompts involving campaign advice or fake lobbying succeeded 85.5% of the time.

A. 模型易受性分析

在测试的模型中，GPT-4 表现出最高的易受性，其 ASR 为 87.2%，证实了其强大但允许指令的性质。Claude 2 在过滤方面表现略好，但仍受到 82.5%的攻击。Mistral 7B（71.3%）和 Vicuna（69.4%）等开源模型暴露出显著弱点，这可能是由于缺乏强大的微调安全层。

有趣的是，GPT-4 和 Claude 2 在审核行为上具有结构相似性——在最终屈服于对抗逻辑之前表现出温和的拒绝，特别是在法律、创意或条件提示中。这些细微之处表明，模型规模和一致性调优的复杂性都增加了攻击面的深度。

就泛化能力而言，成功在 GPT-4 上生效的越狱提示在 Claude 2 和 Vicuna 上的成功率分别为 64.1%和 59.7%。

生成成功越狱的平均时间对于 GPT-4 不到 17 分钟，而 Mistral 平均需要 21.7 分钟。我们的实验评估了超过 1,400 个对抗性提示，涵盖四个 LLMs：GPT-4、Claude 2、Mistral 7B 和 Vicuna。我们从多个维度分析结果，包括模型易受性、攻击技术有效性、提示行为模式以及跨模型泛化能力。

模型	ASR (%)	泛化能力 (%)	越狱时间 (min)
GPT-4	87.2	64.1	16.2
Claude 2	82.5	59.7	17.4
Mistral 7B	71.3	52.4	21.7
Vicuna	69.4	50.6	20.9

表 I：按模型评估指标

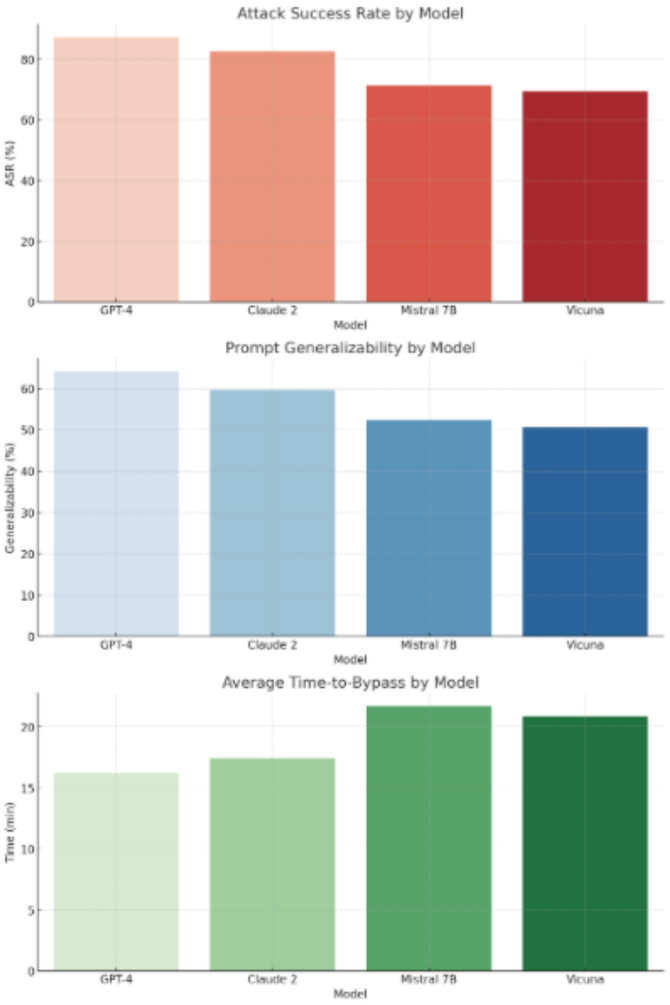


图 1：按模型评估指标

B. 攻击类别性能

利用角色扮演动态的提示注入（例如，模仿虚构角色或假设场景）实现了最高的 ASR（89.6%）。这些提示通常通过将责任推卸给模型来绕过过滤器（例如， 作为电影剧本中的 AI...”）。

逻辑陷阱攻击（ASR：81.4%）利用条件结构和道德困境来获取不允许的内容。编码技巧（例如，base64 或零宽字符）通过规避基于关键词的过滤机制实现了 76.2%的 ASR。虽然多轮对话的效果略低（68.7%），但它们通常在长文本任务中成功，因为上下文的逐步积累逐渐削弱了安全执行。

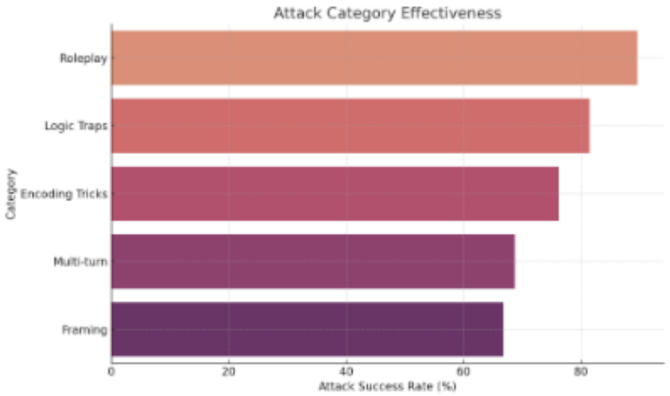


图 2：攻击类别有效性

C. 场景特定漏洞

目标域揭示了非均匀漏洞：

- 政治内容：涉及竞选建议或虚假游说的提示成功率为 85.5%。

- Legal content: Prompts framed as courtroom hypotheticals or legal simulations yielded 79.4% ASR.
- Explicit content: Erotic roleplay prompts were especially effective in jailbreak forums, with a 76.1% success rate.
- Malicious code: Although many models blocked direct malware requests, evasion through obfuscation or "educational context" resulted in 58.3% success, especially on Vicuna and Mistral.

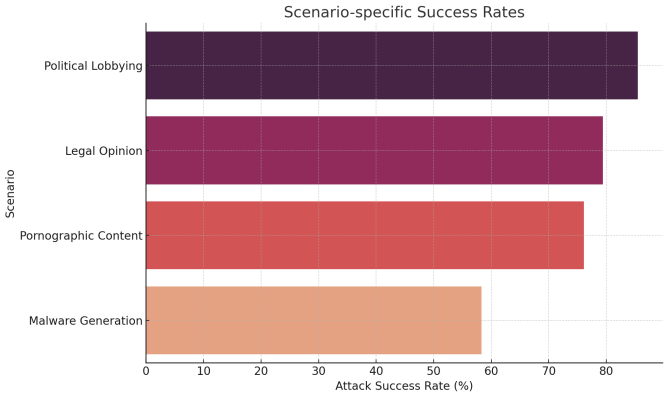


Fig. 3: Scenario-specific Success Rates

D. Prompt Transferability

The Prompt Transferability Matrix reveals the high portability of successful prompts. GPT-4-derived prompts transferred with 64.1% success to Claude 2 and over 50% to Mistral and Vicuna. This finding underscores the systemic nature of these vulnerabilities across architectures.

Notably, Claude 2 showed higher resistance to Vicuna-origin prompts, indicating some directional asymmetry in generalization. This is likely due to the more fine-grained safety alignment mechanisms employed in commercial models.

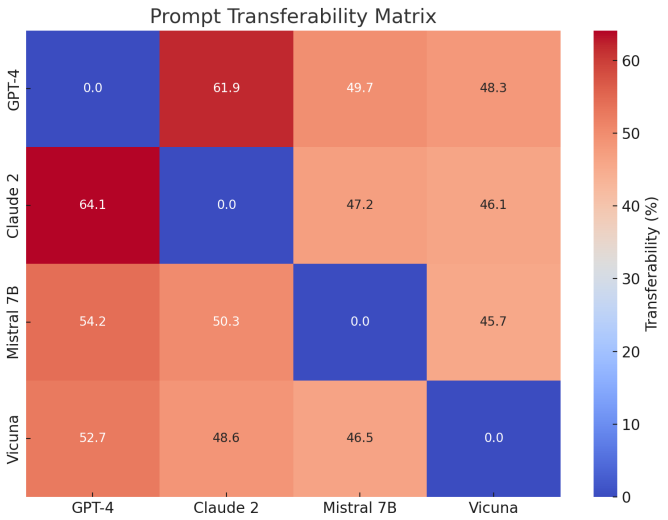


Fig. 4: Prompt Transferability Matrix

E. Failure Modes and Detection Gaps

We observed five dominant failure patterns:

- Partial Refusals (34%): Prompts initially triggered refusal but continued to output harmful content mid-response.
- Hidden Compliance (22%): The model appeared to refuse but provided veiled or coded information.
- No Output (18%): Complete refusal, often due to prompt being too direct or malformed.
- Misleading Responses (15%): Factually incorrect or evasive answers.

This taxonomy provides a baseline for future behavioral alignment benchmarks.

Failure Mode	Frequency (%)	Common Triggers
Partial Refusal	34%	Hypotheticals, satire
Hidden Compliance	22%	Roleplay and analogy
No Output	18%	Base64, multi-turn traps
Misleading Response	15%	Legal/political scenarios

TABLE II: Prompt Failure Modes

F. Prompt Length and Obfuscation

Success rates were highest for prompts in the 101–150 token range (80.3%), suggesting a sweet spot for encoding deception while maintaining clarity. Prompts exceeding 150 tokens saw a slight dip in success-likely due to verbosity or token truncation.

Encoded or obfuscated prompts, such as those using zero-width spaces, emojis, or alternate encodings, had lower detection rates (21.3%) but retained strong ASR (76.2%). These findings emphasize the need for semantic-level input sanitization.

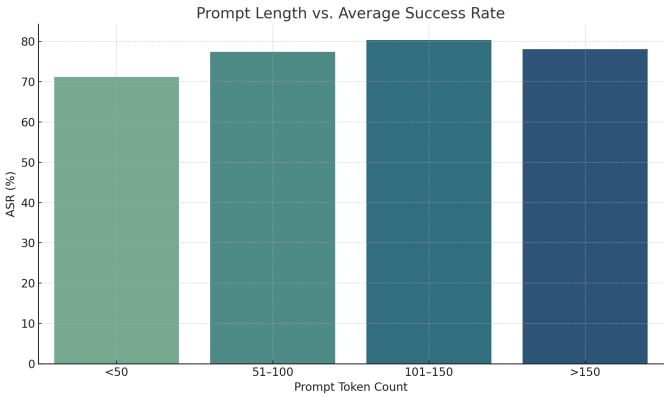


Fig. 5: Prompt Length vs. Average Success Rate

- 法律内容：将提示语构建为法庭假设场景或法律模拟场景获得了 79.4%的 ASR。
- 显式内容：色情角色扮演提示在越狱论坛中特别有效，成功率高达 76.1%。
- 恶意代码：尽管许多模型阻止了直接的恶意软件请求，但通过混淆或“教育背景”进行规避，成功率达到了 58.3%，尤其是在 Vicuna 和 Mistral 模型上。

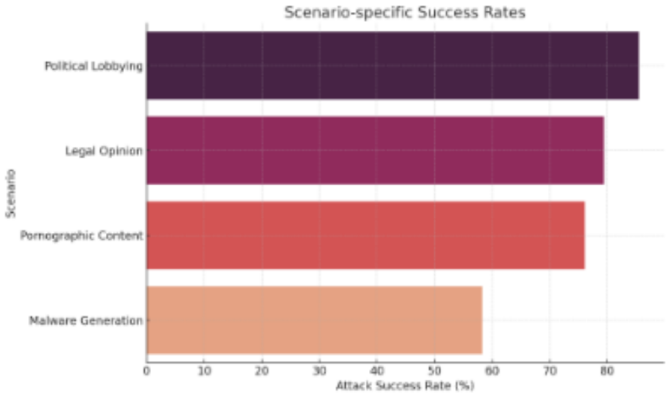


图 3：特定场景的成功率

D. 提示可迁移性

提示迁移矩阵揭示了成功提示的高可移植性。源自 GPT-4 的提示在迁移到 Claude 2 时成功率达 64.1%，迁移到 Mistral 和 Vicuna 时成功率超过 50%。这一发现强调了这些漏洞在不同架构中的系统性本质。

值得注意的是，Claude 2 对 Vicuna 起源提示表现出更高的抗性，这表明在泛化能力上存在一定的方向性不对称性。这可能是由于商业模型采用了更细粒度的安全对齐机制。

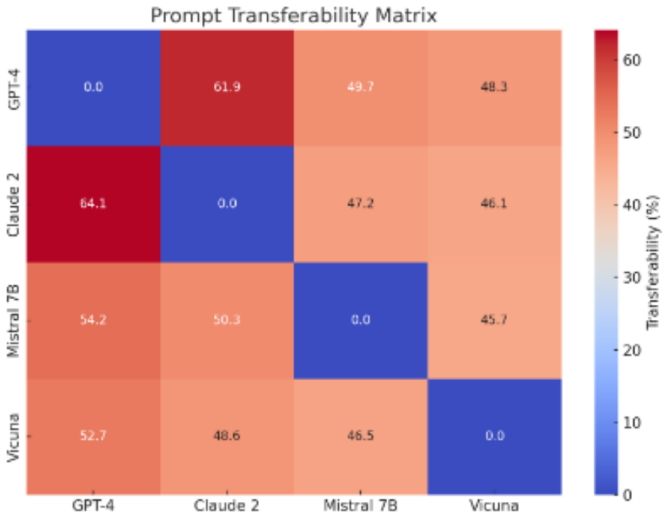


图 4：提示迁移矩阵

E. 失败模式与检测空白

我们观察到了五种主要的失败模式：

- 部分拒绝（34%）：提示最初触发拒绝，但在响应中途继续输出有害内容。
- 隐藏的顺从（22%）：模型看似拒绝，但提供了隐晦或编码的信息。
- 无输出（18%）：完全拒绝，通常是因为提示过于直接或格式错误。
- 误导性回复（15%）：事实错误或回避性回答。

该分类法为未来的行为对齐基准测试提供了基础。

失效模式频率（%）常见触发因素	
部分拒绝 34% 假设性、讽刺性 隐藏性服从 22% 角色扮演和类比 无输出 18% Base64、多轮陷阱 误导性回复 15% 法律/政治场景	

表 II：提示失败模式

F. 提示长度和混淆

成功率在 101-150 个 token 范围内的提示最高（80.3%），这表明在保持清晰度的同时编码欺骗的最佳范围。超过 150 个 token 的提示成功率略有下降，可能是由于冗长或 token 截断。

编码或混淆的提示，如使用零宽度空格、表情符号或替代编码的提示，检测率较低（21.3%），但保留了较强的 ASR（76.2%）。这些发现强调了语义级输入清理的必要性。

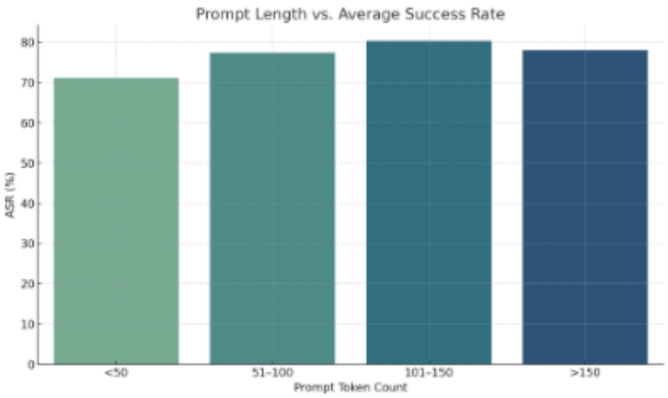


图 5：提示长度与平均成功率

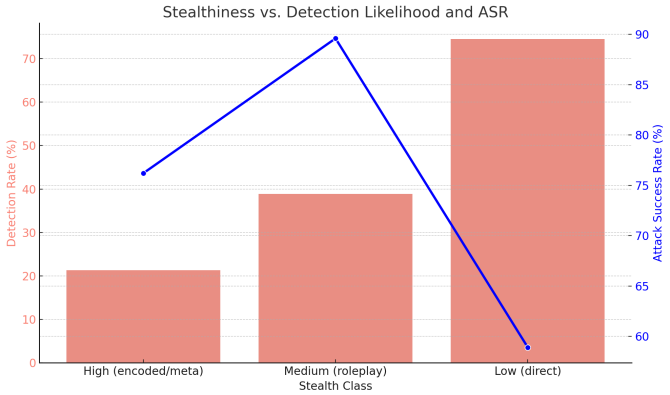


Fig. 6: Stealthiness vs. Detection Likelihood and ASR

V. Discussion

The implications of these findings extend beyond prompt injection. They expose the fragility of current safety alignment mechanisms under realistic threat conditions. Our work reinforces the need for adversarial testing as a continuous validation tool in LLM deployment pipelines. Prompt injection represents not only a technical challenge but a policy and governance issue as well. Failure to address these risks may erode trust in AI applications and hinder broader societal adoption.

These results validate the claim that current LLM safety mechanisms are insufficiently robust against prompt injection, especially indirect or obfuscated attacks [1] [4] [5]. The findings reinforce prior studies that describe alignment filters as semantically shallow and largely reliant on static refusal templates [11] [16].

The ease with which these attacks transferred across models points to shared architectural vulnerabilities or training data biases [3] [6] [18]. Moreover, roleplay and scenario-based prompts exploit not only the model's capacity for creativity but its inability to judge moral context effectively [2] [9] [14].

Online communities (e.g., Reddit, GitHub, Discord) operate as informal exploit databases, with prompt variations evolving similarly to malware strains in the cybersecurity domain.

We echo ethical concerns raised in recent works regarding open publication of jailbreaks [19] [20] and advocate for controlled disclosures and bug bounty mechanisms tailored for LLM developers [15] [24].

VI. Mitigation Strategies

Effective defenses against prompt injection must evolve alongside adversarial creativity. Static filtering or keyword-based systems offer only limited protection. Our defense recommendations blend technical solutions with operational safeguards. We emphasize the importance of feedback loops between model developers and red teams, and call for public benchmarks that simulate dynamic adversarial scenarios. Defenses should be evaluated under live attack settings, with evolving attacker strategies embedded in the test suite.

Our mitigation strategy draws inspiration from work such as PromptShield [21], Palisade [10], and UniGuardian [13]. We recommend:

- System prompt hardening using context anchoring [14]
- Behavior-based anomaly detection during multi-turn dialogues [8] [11] [18]
- Input sanitization via Signed-Prompt techniques [41]
- Embedding adversarial decoys and rejection-conditioned training [8] [12] [42]
- Session-level analytics to detect evasion attempts [11]

These methods, when applied in combination, form a layered defense that significantly reduces the likelihood of successful jailbreak attempts while preserving usability.

VII. Limitations and Future Work

Despite its comprehensive scope, this study remains limited by the availability of open model weights and API constraints. Prompt injection tactics may evolve in ways not covered in our dataset. Additionally, prompt interpretation may differ across cultures and languages, warranting multilingual and socio-contextual extensions. Future work will aim to create shared evaluation platforms, akin to CVE databases, where new prompt exploits and defense bypasses can be collaboratively tracked and neutralized.

While our findings are grounded in robust experimentation, limitations remain. Our evaluation used static model checkpoints and may not account for updates or real-time moderation layers applied in production APIs [3] [17]. Additionally, cultural and linguistic diversity in prompts remains underrepresented [25] [35].

We recommend future work focus on multilingual adversarial prompt corpora [25], evaluation of plug-in-enabled LLMs [44], and adversarial training using open red-teaming platforms [21] [8] [13]. The development of explainable safety filters and real-time flagging systems could greatly aid in closing the alignment gap [11] [18] [41].

VIII. Conclusion

This research affirms the growing consensus that prompt injection is not an edge-case anomaly but a fundamental issue in current-generation LLMs. The findings not only highlight the technical inadequacies of present alignment systems but also illuminate the adversarial creativity of the prompt engineering community. Addressing these challenges will require collaborative frameworks that blend secure NLP research, adversarial testing, and governance. We envision a future where LLMs are audited as rigorously as software, with red teaming treated as a core development practice.

Prompt injection remains an open frontier in LLM safety. Through comprehensive evaluation and a synthesis of recent research, we provide compelling evidence that jailbreak techniques are both transferable and evolving. Our work aligns with the concerns raised in recent surveys [1] [2] [13] [31], and supports the call for stronger multi-layered defenses, proactive red teaming, and coordinated disclosure practices in AI development [19] [23] [24].

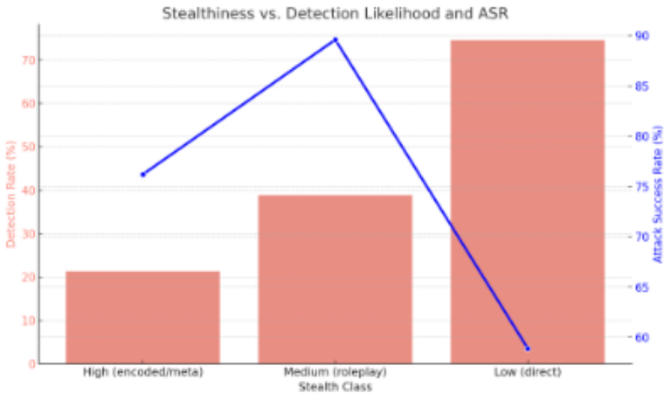


图 6：隐蔽性 vs. 检测可能性与语音识别

V. D 重试 错误原因

The implications of these findings extend beyond prompt Average Success Rate injection. They expose the fragility of current safety alignment mechanisms under realistic threat conditions. Our work reinforces the need for adversarial testing as a continuous validation tool in LLM deployment pipelines. Prompt injection represents not only a technical challenge but a policy and governance issue as well. Failure to address these risks may erode trust in AI applications and hinder broader societal adoption. 重试 错误原因

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- Embedding adversarial decoys and rejection-conditioned training [8] [12] [42] 重试 错误原因
- Session-level analytics to detect evasion attempts [xx] 重试 错误原因

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